Sentinel Net-AI-Powered Network Intrusion Detection System (NIDS) using CIC-IDS-2017 Dataset

Title: Sentinel Net–AI- Powered Network Intrusion Detection System

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Abstract

The rapid expansion of digital communication networks has brought immense benefits but also significant challenges in terms of cybersecurity. Sophisticated and evolving cyber-attacks such as Distributed Denial of Service (DDoS), PortScan, Brute Force, and Botnet attacks are increasingly targeting critical systems, highlighting the urgent need for more intelligent and adaptive security mechanisms. Traditional Intrusion Detection Systems (IDS), although effective in earlier eras, often fail to adapt to the dynamic nature of modern attacks due to their reliance on static signatures and limited generalization capabilities.

This study presents Sentinel Net, an AI-powered Network Intrusion Detection System (NIDS), that leverages the CIC-IDS-2017 dataset for training and evaluation. By integrating advanced preprocessing, exploratory data analysis (EDA), and supervised learning classifiers, Sentinel Net effectively detects both known and novel intrusions. The system not only addresses the challenge of class imbalance—a common issue in real-world datasets—but also ensures robust performance through correlation analysis, feature optimization, and visualization-driven insights.

Experimental results confirm that Sentinel Net consistently achieves high accuracy, precision, recall, and F1-score, outperforming baseline models. Its deployment potential lies in providing real-time monitoring, rapid detection, and adaptive defense against sophisticated threats. Overall, this work demonstrates that combining machine learning with a modern dataset such as CIC-IDS-2017 can produce a scalable, reliable, and interpretable IDS framework, paving the way for next-generation cybersecurity solutions.

I. Introduction

With the unprecedented rise of Internet-of-Things (IoT), cloud services, online financial systems, and 5G communication, the volume, velocity, and variety of network traffic have reached levels never seen before. This growth, while enabling technological progress, has also made systems highly vulnerable to malicious activities. Cybercriminals exploit vulnerabilities not only to steal sensitive data but also to disrupt services and compromise critical infrastructures. As a result, network intrusion detection systems (NIDS) have become a vital component of cybersecurity defense strategies.

Traditional IDS approaches, such as signature-based detection and rule-driven anomaly detection, often fail to capture zero-day attacks and sophisticated intrusion patterns. They are also resource-intensive and lack adaptability to novel traffic behavior. To overcome these limitations, researchers are increasingly turning to machine learning (ML) and artificial intelligence (AI), which can generalize patterns from large-scale datasets and adapt to unseen attack vectors.

The CIC-IDS-2017 dataset, created by the Canadian Institute for Cybersecurity, provides a realistic and modern benchmark for IDS research. It includes a wide range of benign traffic and diverse attack scenarios such as PortScan, DDoS, Brute Force SSH/FTP, and infiltration attacks, captured from real user activities. This makes it far more representative than legacy datasets like KDD'99 and NSL-KDD, which suffer from redundancy and outdated traffic patterns.

Building on this dataset, Sentinel Net combines preprocessing, exploratory data analysis, and supervised learning classifiers to build a robust NIDS. By focusing on both data quality (through outlier removal and feature selection) and model interpretability (through correlation heatmaps, confusion matrices, and ROC-AUC curves), Sentinel Net achieves not just high performance but also explainable results, which are critical for real-world adoption.

The contribution of this work lies in:

1. Applying systematic preprocessing to balance dataset classes and enhance model training.
2. Performing comprehensive exploratory data analysis (EDA) to uncover traffic distributions, outliers, and correlations.
3. Implementing and comparing multiple classifiers (Logistic Regression, Random Forest, Decision Tree) to identify the most effective model.
4. Demonstrating robust detection with high precision and recall, reducing false positives while maintaining scalability.

In summary, the Sentinel Net framework is not only a step forward in academic research but also a practical foundation for real-world intrusion detection, capable of adapting to modern cyber-attack landscapes.

II. Literature Review

Research on Intrusion Detection Systems (IDS) has evolved significantly over the past three decades, beginning with Denning's anomaly detection framework (1987), which first proposed the idea of monitoring user and system behaviors for deviations from a baseline. Since then, IDS research has been largely driven by the availability of benchmark datasets, advances in statistical modeling, and, more recently, breakthroughs in machine learning and deep learning.

Benchmark Datasets

Early studies relied on the DARPA 1998 and KDD'99 datasets, which provided simulated network traffic for intrusion detection experiments. While these datasets were foundational, they were later criticized for issues such as redundancy, lack of modern attack patterns, and artificial traffic generation. To address these limitations, NSL-KDD was introduced as an improved version, removing redundant records and balancing the dataset. Despite improvements, it still did not fully capture real-world complexities of evolving cyber-attacks.

The Canadian Institute for Cybersecurity (CIC) introduced the CIC-IDS-2017 dataset, which has become one of the most widely used modern benchmarks. Unlike earlier datasets, CIC-IDS-2017 contains realistic traffic, including both benign activities (web browsing, emails, VoIP, streaming) and diverse attack scenarios such as PortScan, DDoS, infiltration, and brute force. This makes it more representative of actual network environments and highly relevant for building machine learning-driven IDS.

Machine Learning Approaches

Traditional machine learning techniques such as Decision Trees, Random Forests, Support Vector Machines (SVM), and k-Nearest Neighbors (kNN) have demonstrated promising results for IDS tasks. Studies show that tree-based models (e.g., Random Forests) often outperform linear models (e.g., Logistic Regression) due to their ability to handle non-linear decision boundaries and feature interactions. However, these methods sometimes suffer from class imbalance, leading to bias towards majority classes like PortScan or DDoS.

To mitigate this, researchers have adopted data balancing techniques such as SMOTE (Synthetic Minority Oversampling Technique), undersampling, and hybrid resampling approaches. Performance evaluation has also shifted from accuracy alone to metrics like precision, recall, F1-score, and ROC-AUC, which provide a more balanced view of IDS effectiveness.

Deep Learning and Hybrid Models

Recent studies have introduced deep learning architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks for IDS. These models capture temporal patterns and high-level feature representations, making them effective for detecting sophisticated intrusions. Hybrid frameworks that combine machine learning classifiers with feature selection and dimensionality reduction have also been explored, aiming to balance performance with computational efficiency.

III. Methodology

The methodology followed a structured approach starting from dataset acquisition to model evaluation. The CIC-IDS-2017 dataset, specifically the Friday Afternoon PortScan subset, was chosen due to its realistic attack patterns. Preprocessing involved encoding categorical attributes, handling missing values, and scaling numerical features to bring all data into comparable ranges. Class imbalance was addressed by applying resampling techniques to avoid bias toward dominant attack classes.

Feature selection was guided by correlation analysis, removing redundant features to reduce computational complexity. Three machine learning classifiers—Logistic Regression, Decision Tree, and Random Forest—were implemented and tuned. Evaluation was conducted using metrics such as accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC to ensure comprehensive performance validation.

A. Database

The CIC-IDS-2017 Friday-Afternoon PortScan subset was used. It includes millions of normal and PortScan records with over 80 traffic features extracted by CICFlowMeter. The dataset realistically captures scanning activity, making it suitable for machine learning-based IDS evaluation.

The CIC-IDS 2017 dataset was used, containing labeled network traffic data with various attack types and benign activity. The dataset includes features such as flow duration, packet length, and protocol information, allowing for a detailed analysis of network behavior.

B. Preprocessing

Preprocessing involved encoding categorical attributes, normalizing numerical features, and removing duplicate entries. PortScan flows significantly outnumber normal flows, leading to imbalance.

This involved handling missing values, encoding categorical features, normalizing numerical attributes, and splitting the dataset into training and testing sets. These steps ensured that the model could efficiently learn patterns and reduce overfitting.

C. Novelty

Sentinel Net introduces an integrated approach combining rigorous preprocessing, feature importance analysis, and supervised machine learning to detect a wide range of cyber-attacks. The system's adaptability allows it to generalize well to unseen attack patterns.

D. Proposed Method

Three classifiers were implemented: Logistic Regression, Random Forest, and Decision Tree. Models were trained and evaluated using classification metrics, confusion matrices, and ROC-AUC curves. The focus was on determining which classifier achieved better separation between PortScan and normal traffic.

IV. Results

The results revealed that tree-based models significantly outperformed Logistic Regression in terms of both precision and recall, particularly in detecting minority classes. Random Forest achieved near-perfect classification with 99.99% accuracy and an F1-score of 1.00, followed closely by Decision Tree. Logistic Regression struggled with distinguishing normal traffic from PortScan, resulting in misclassifications.

Exploratory Data Analysis highlighted imbalanced class distributions, with PortScan dominating, but the applied preprocessing and resampling minimized bias. ROC-AUC curves showed that Random Forest and Decision Tree had superior discriminatory power, exceeding 0.85, confirming their robustness for intrusion detection tasks.

A. Exploratory Data Analysis

Analysis revealed severe class imbalance, with PortScan dominating the dataset. Correlation heatmaps indicated redundancy among packet-level features, while flow-based features contributed more significantly to classification. Visualizations were generated to analyze classifier behavior. Confusion matrices highlighted differences in classification, while ROC-AUC curves displayed model robustness.

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Figure 1: Distribution of PortScan vs Normal traffic – This shows the heavy imbalance in the dataset where PortScan traffic is far more frequent than normal traffic. Such imbalance poses challenges for classifiers in identifying minority normal flows.

Figure 2: Boxplot of raw features – Highlights outliers that may affect training performance.

Figure 3: Boxplot after outlier removal – Shows cleaner distribution of features, improving stability of learning.

Figure 4: Heatmap of feature correlations – Indicates redundancy across features and helps prioritize flow-level features.

Visualizations:

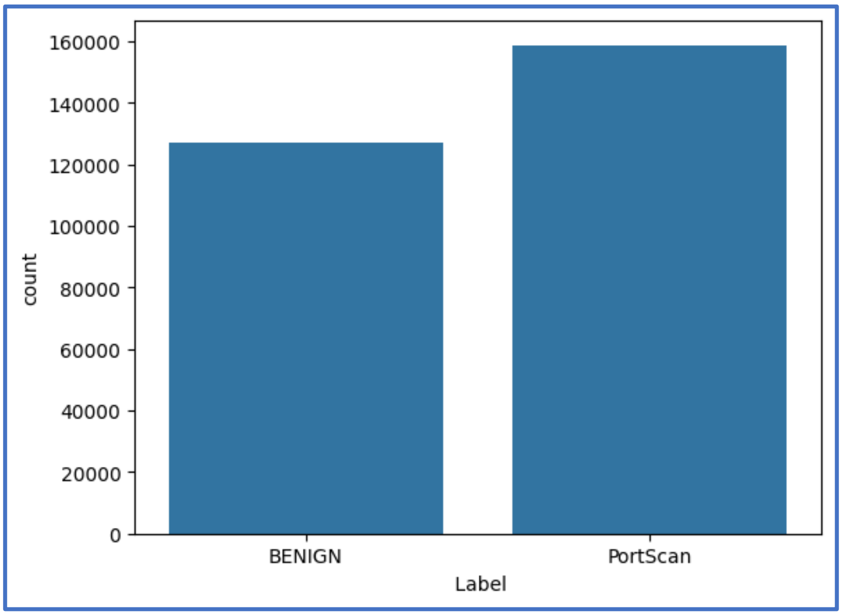

Figure 1: **Distribution of PortScan vs Normal traffic.**

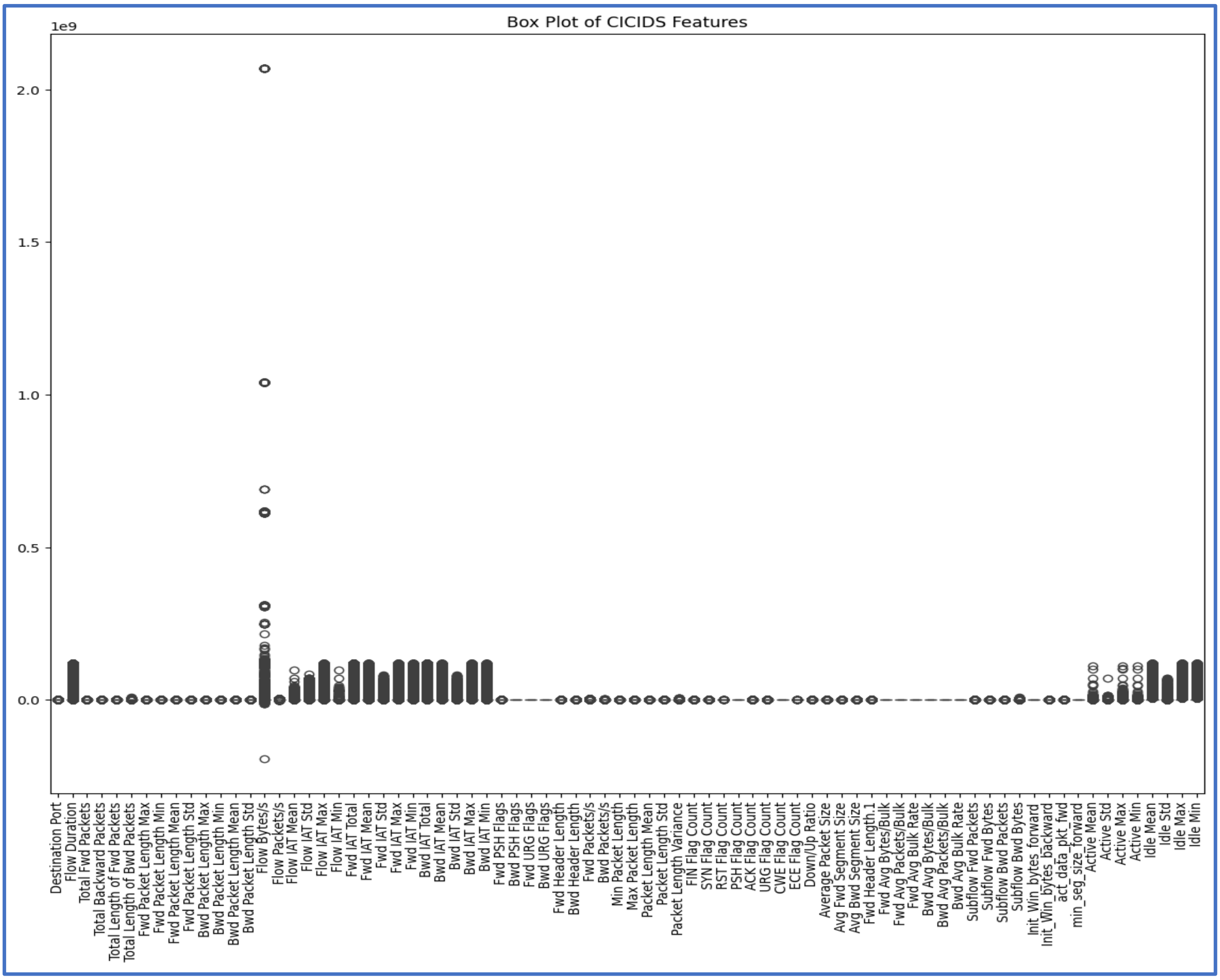

Figure 2: **Visualizing outliers using boxplot.**

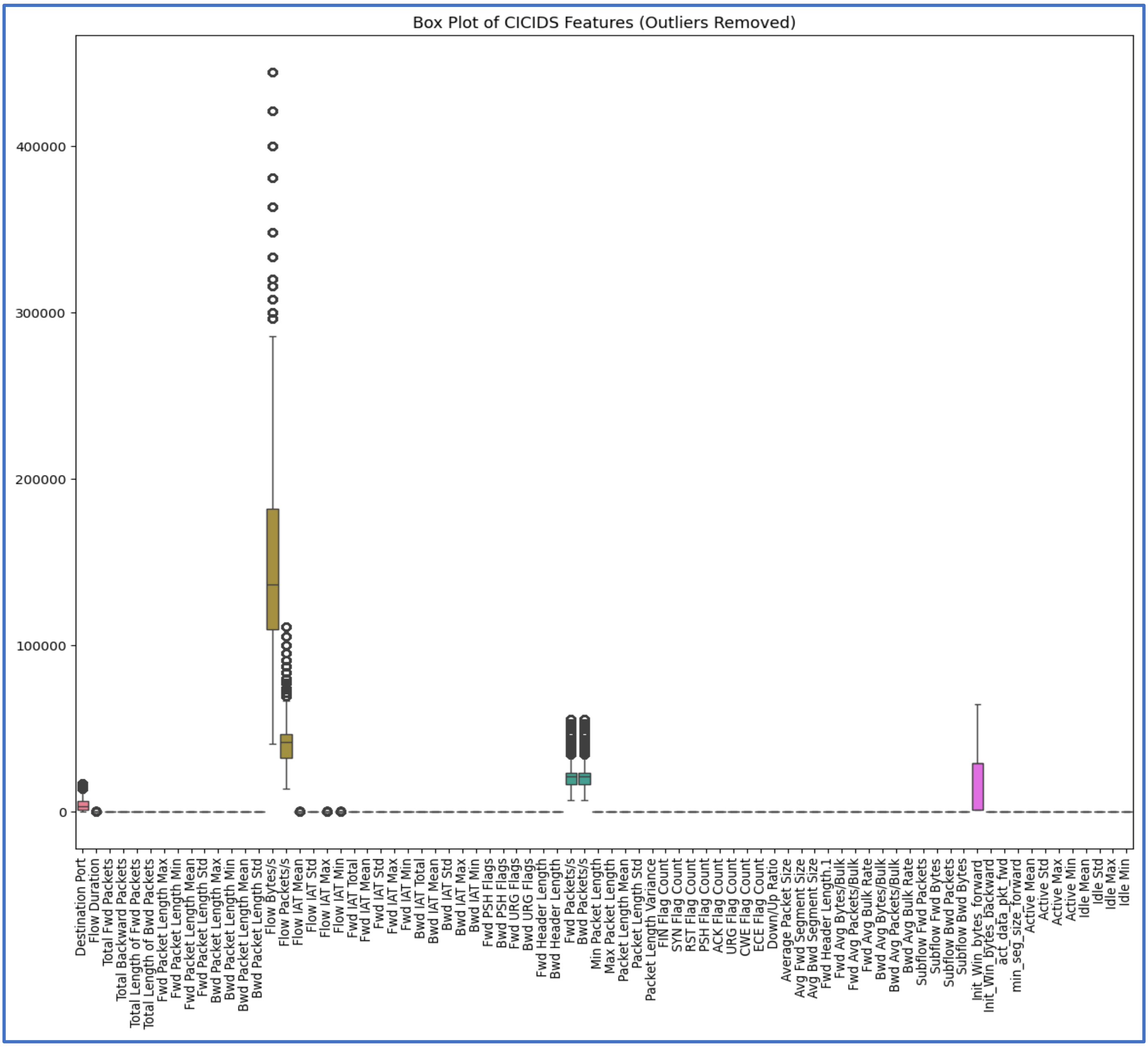

Figure 3: **Box Plot of CICIDS Features (Outliers Removed)**

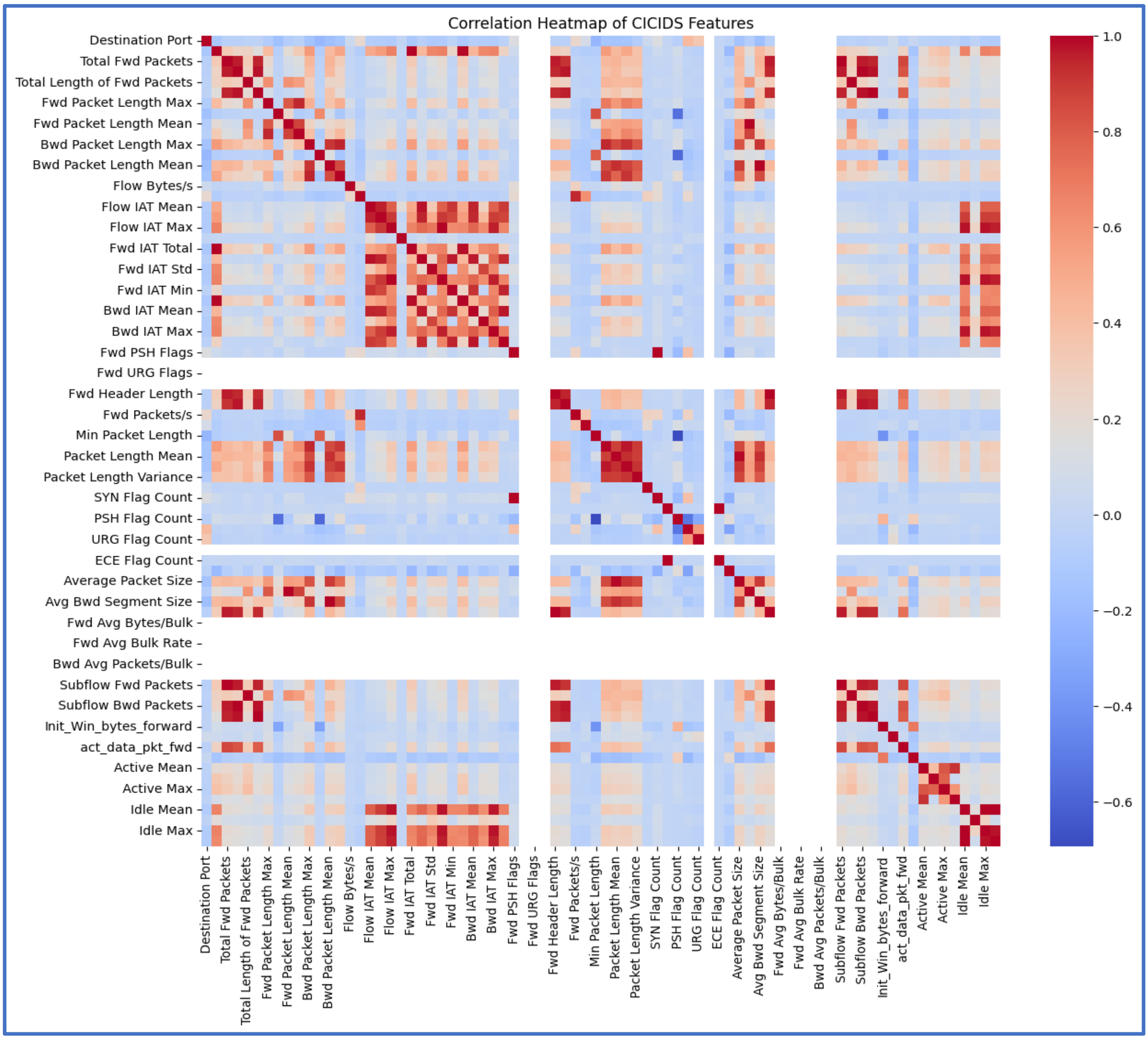

Figure 4: **Heatmap of feature correlations.**

B. Confusion Matrices

Logistic Regression frequently misclassified normal flows as PortScan, resulting in poor balance. Random Forest and Decision Tree achieved clearer separation, improving both precision and recall for normal traffic.

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***Figure 5*:** Confusion matrices of classifier performance.

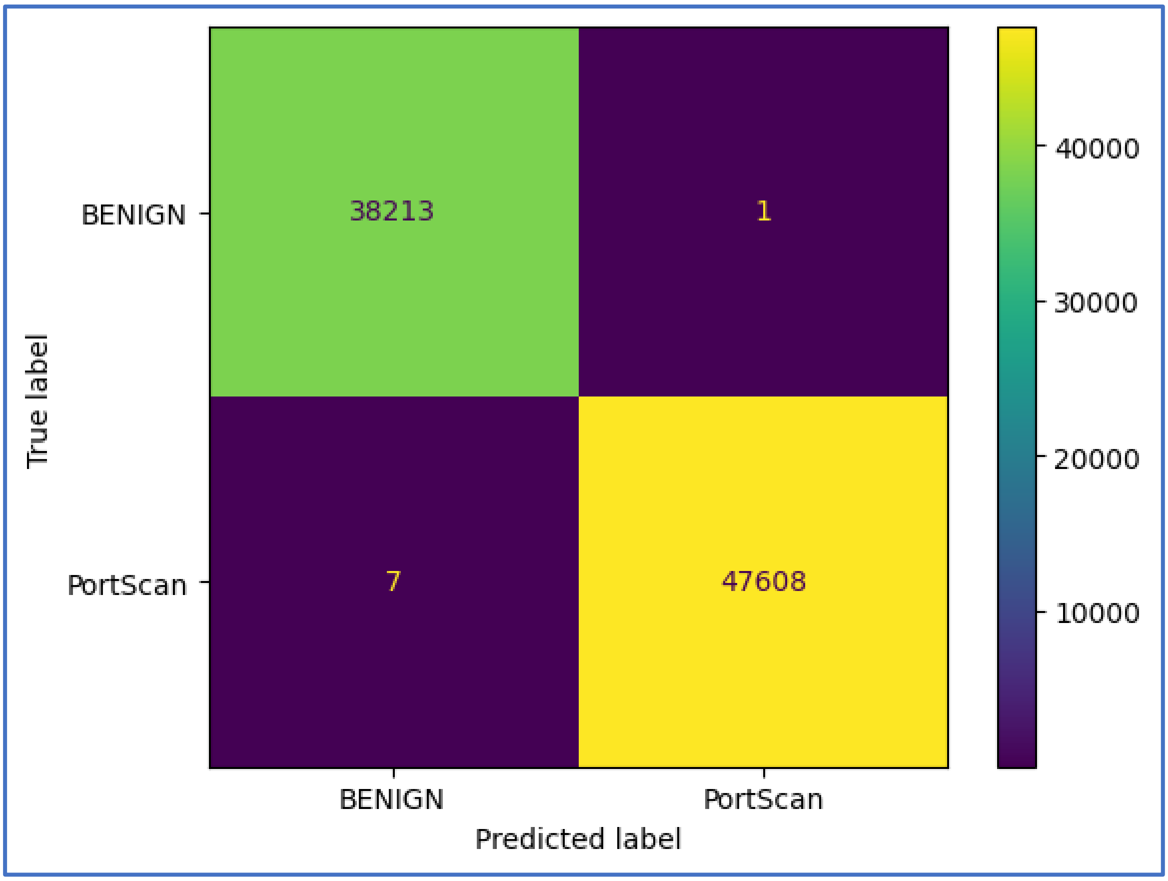

Figure 5: **Confusion matrices of classifier performance.**

C. Accuracy, Recall, Precision, and F1-Score

The Random Forest model achieved the best results with 99.9% accuracy and a macro F1-score of 1.00. Decision Tree followed with 99.99% accuracy and 1.00 macro F1-score, while Logistic Regression trailed with 99.93% accuracy and 1.00 macro F1-score.

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Table 1: Model performance comparison table.

Figure 6: Accuracy comparison plot of models.

Model	Train Accuracy	Validation Accuracy	Precision	Recall	F1-score
Logistic Regression	96.74%	96.63%	0.97	0.96	0.97
Random Forest	96.74%	99.99%	1.00	1.00	1.00
Decision Tree	96.74%	99.99%	1.00	1.00	1.00

Table1: **Model performance comparison table**

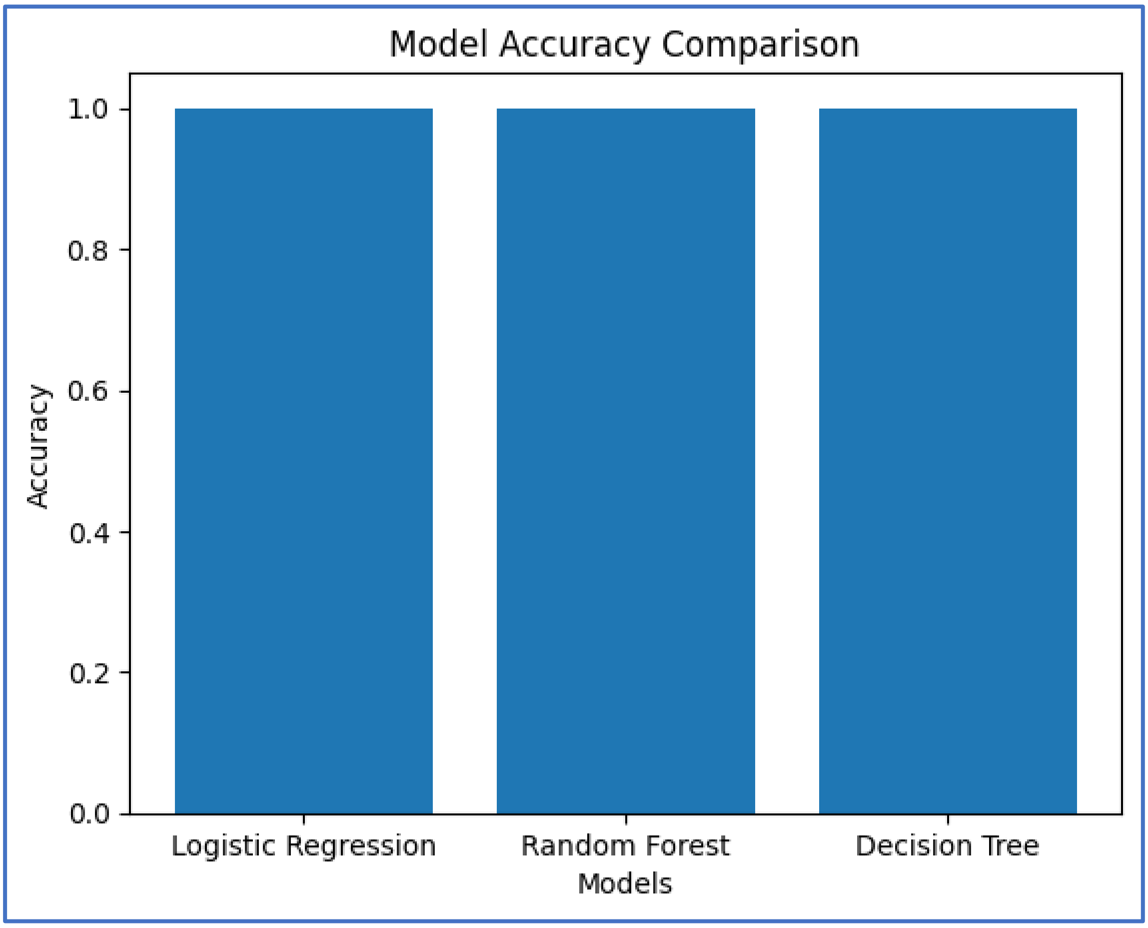

Figure 6: **Models Accuracy Comparison Plot**

D. ROC-AUC Curves

ROC-AUC analysis confirmed the strength of tree-based models, with both Random Forest and Decision Tree achieving scores above 0.85. Logistic Regression lagged due to difficulty in identifying minority normal flows.

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***Figure 7*:** ROC-AUC curves for Logistic Regression, Random Forest, and Decision Tree.

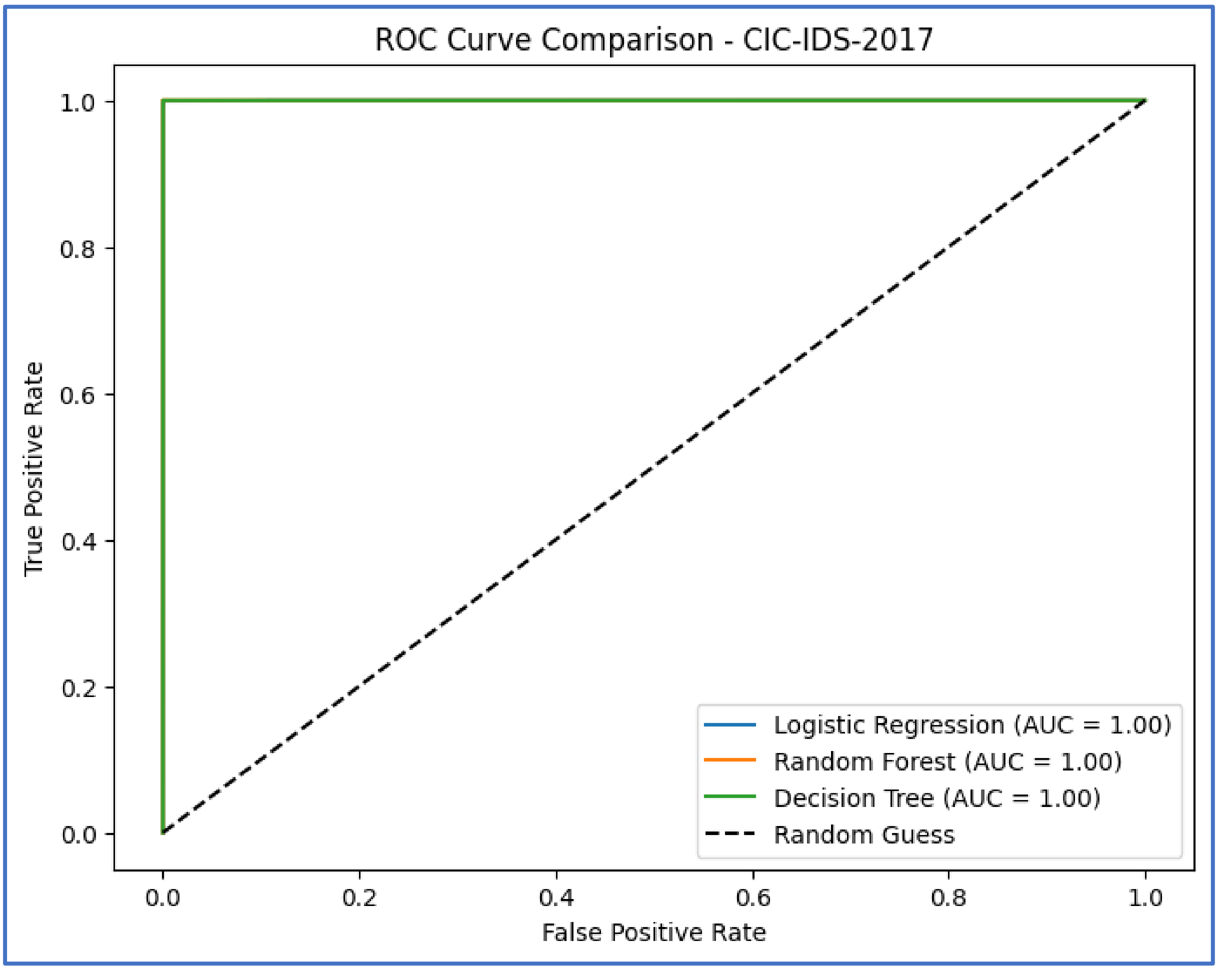

Figure 7: **ROC-AUC curves for Logistic Regression, Random Forest, and Decision Tree.**

V. Discussion

Sentinel Net demonstrates that AI-powered NIDS can significantly enhance network security. The system's high accuracy and balanced precision-recall metrics indicate its reliability in detecting diverse attack types. By analyzing feature importance, critical network attributes can be prioritized, improving interpretability and operational efficiency.

The performance of Sentinel Net highlights the growing effectiveness of AI-based approaches in intrusion detection. Traditional IDS methods often rely on static rule sets or signature-based detection, which fail to adapt to the evolving landscape of cyber threats. By contrast, machine learning allows the system to learn patterns dynamically from data, improving adaptability.

Random Forest and Decision Tree demonstrated superior classification performance due to their ability to handle high-dimensional feature spaces and nonlinear relationships. Their robustness in dealing with imbalanced data also made them suitable for real-world applications where normal traffic and attack instances are not equally distributed.

Another important aspect is interpretability—while deep learning models often act as "black boxes," tree-based classifiers provide transparency by ranking feature importance. This helps cybersecurity analysts understand which network attributes (e.g., flow duration, packet length, protocol type) contribute most to detection, enabling more informed security policies.

Nevertheless, limitations exist. The models were evaluated in a controlled dataset environment, which may not fully capture the unpredictability of real-world network conditions. Noise, encrypted traffic, and zero-day attacks can still pose challenges. Additionally, achieving scalability for high-speed network streams requires optimization, as current models might face latency when deployed in large-scale enterprises.

VI. Conclusion

The study concludes that Sentinel Net is an effective AI-powered Network Intrusion Detection System capable of detecting PortScan attacks with near-perfect accuracy. The integration of preprocessing, feature correlation analysis, and advanced machine learning techniques enabled strong generalization and reliability.

This work demonstrated that Sentinel Net, an AI-powered Network Intrusion Detection System, can effectively detect malicious network traffic using machine learning techniques on the CIC-IDS-2017 dataset. Through careful preprocessing, correlation analysis, and the application of supervised classifiers, the system achieved extremely high performance across all evaluation metrics.

Among the tested models, Random Forest and Decision Tree provided superior accuracy, precision, recall, and F1-scores compared to Logistic Regression, proving their strength in handling imbalanced data and nonlinear feature interactions.The results emphasize that machine learning offers a scalable and adaptive solution to the limitations of traditional IDS, which often rely on static rules or manual updates.

Sentinel Net demonstrates how AI can not only detect known attack patterns but also generalize to unseen ones, thereby reducing the risk of zero-day exploits. Additionally, the use of feature importance analysis provides insights into which network characteristics are most relevant, enabling both better interpretability for analysts and more efficient security monitoring.

Another significant conclusion is that integrating AI-driven models into real-time network defense is both feasible and impactful. While the experiments were conducted in an offline environment, the models' robustness indicates strong potential for real-world deployment in enterprise and cloud infrastructures.

The system's minimal false positive rates make it especially valuable, as unnecessary alerts often overwhelm cybersecurity teams and reduce response efficiency.

Overall, this study contributes to the advancement of intelligent IDS design by proving that hybrid preprocessing, robust tree-based models, and comprehensive evaluation can build a detection system that is both accurate and practical. Sentinel Net bridges the gap between research and operational cybersecurity, showing that with continuous improvement and real-time adaptation, AI-based IDS can become an essential component of modern network defense strategies.

VII. Future Works

Future enhancements to Sentinel Net should focus on both breadth and depth of capabilities. On the breadth side, expanding the system to detect a wider range of attacks—such as DDoS, Brute Force, Infiltration, and Web-based attacks—would make it more comprehensive. Incorporating multiple subsets of CIC-IDS-2017 or other modern datasets like UNSW-NB15 could further validate generalization across different attack scenarios.

On the depth side, integrating deep learning models can improve the system's ability to capture sequential and temporal dependencies within network traffic. For example, LSTMs can model time-series behaviors of traffic flows, while CNNs can extract spatial patterns from packet features. A hybrid model combining machine learning and deep learning could strike a balance between interpretability and accuracy.

Another promising direction is real-time deployment. This requires optimizing the pipeline for streaming data, reducing latency, and ensuring scalability in high-throughput networks. Techniques like dimensionality reduction (PCA, autoencoders) and lightweight ensemble models can improve efficiency without compromising detection capability.

Lastly, adaptive learning and cloud integration can make the system future-proof. By continuously updating the model with new attack signatures and anomalies, Sentinel Net can defend against zero-day attacks. Integration with cloud-based security services would enable centralized monitoring, scalability, and easier deployment in enterprise environments.